AGI

Benchmarks
Feasibility
Implications

Frederik Schittny | Institute of Theoretical Informatics, KIT
16.02.2026


Sam Altman
@sama
In from three to eight years we will have a machine with the general intelligence of an average human being
8:11 PM · Mar 31, 2025 · 5.5M Views
2K
4K
58K
3.4K

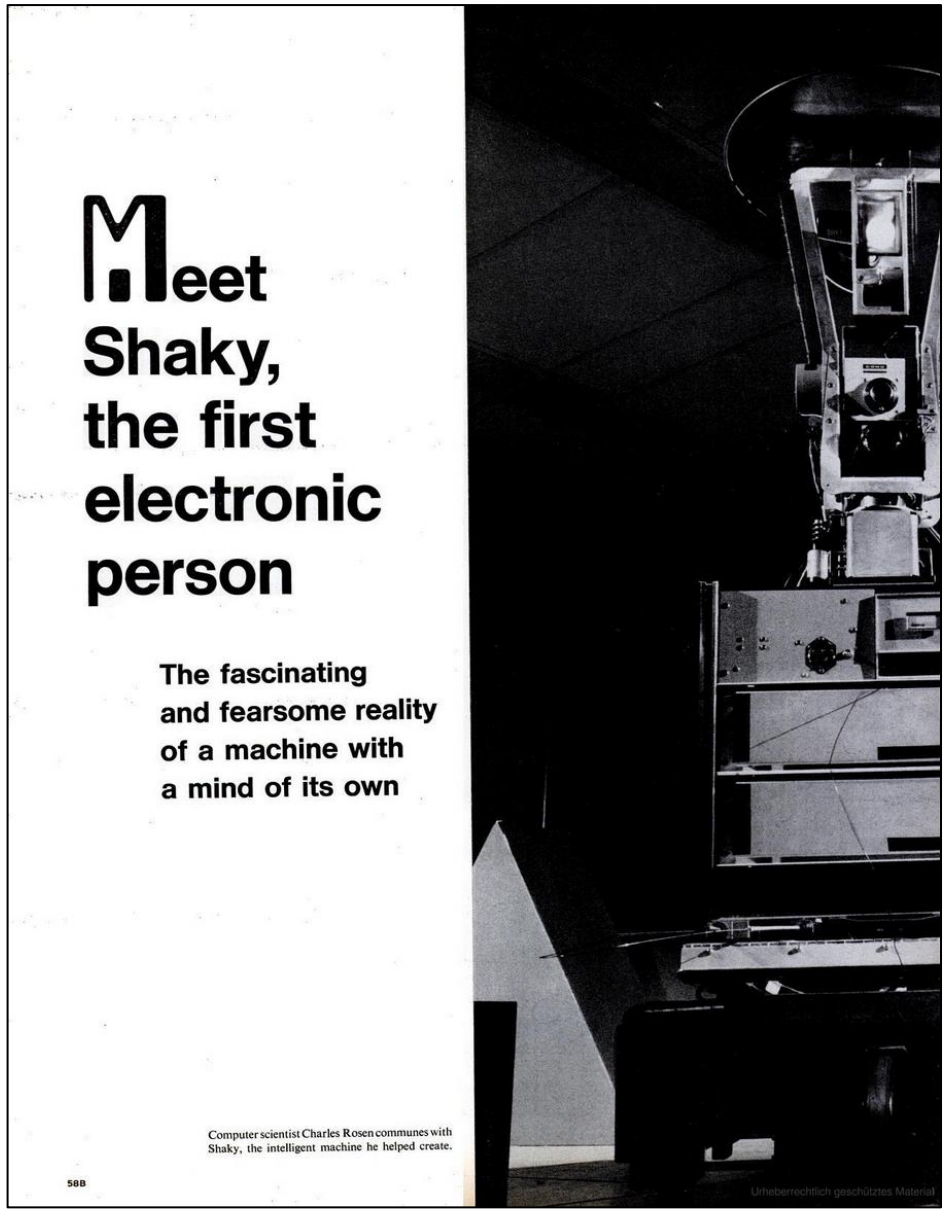

Marvin Minsky
@marv_minsky
In from three to eight years we will have a machine with the general intelligence of an average human being
8:11 PM · Mar 31, 2025 · 5.5M Views
2K
4K
58K
3.4K



"In from three to eight years we will have a machine with the general intelligence of an average human being."

– Marvin Minsky, *Life* Magazine, November 1970

[Life Magazine, 1970]



"In from three to eight years we will have a machine with the general intelligence of an average human being."

– Marvin Minsky, *Life* Magazine, November 1970

[Life Magazine, 1970]

Structure

- AGI Definition – *Ambiguous Foundations*
- Benchmarks – *Some Proposals*
- Feasibility – *Timelines and Approaches*
- Implications – *Propagating Inaccuracy*
- Discussion

AGI

[Russel & Norvig, 2020] [Hendrycks et al., 2025] [Hutson, 2025]

AGI

Artificial

[Russel & Norvig, 2020] [Hendrycks et al., 2025] [Hutson, 2025]

AGI

Artificial General

[Russel & Norvig, 2020] [Hendrycks et al., 2025] [Hutson, 2025]



[Russel & Norvig, 2020] [Hendrycks et al., 2025] [Hutson, 2025]

Defining Intelligence (is really hard)

[Hendrycks et al., 2025] [Hutson, 2025]

Defining Intelligence (is really hard)

internal processes

behavior

[Hendrycks et al., 2025] [Hutson, 2025]

Defining Intelligence (is really hard)

internal processes

behavior

rationality

“human”

[Hendrycks et al., 2025] [Hutson, 2025]

Defining Intelligence (is really hard)

internal processes

behavior

psychology

rationality

“human”

[Hendrycks et al., 2025] [Hutson, 2025]

Defining Intelligence (is really hard)

internal processes

behavior

psychology

neuroscience

rationality

“human”

[Hendrycks et al., 2025] [Hutson, 2025]

Defining Intelligence (is really hard)

philosophy

internal processes

behavior

psychology

neuroscience

rationality

“human”

[Hendrycks et al., 2025] [Hutson, 2025]

Defining Intelligence (is really hard)

philosophy

internal processes

behavior

psychology

neuroscience

rationality

“human”

biology

[Hendrycks et al., 2025] [Hutson, 2025]

Defining Intelligence (is really hard)

philosophy

internal processes

behavior

psychology

neuroscience

rationality

“human”

biology

sociology

[Hendrycks et al., 2025] [Hutson, 2025]

Defining Intelligence (is really hard)

philosophy

internal processes

behavior

mathematics

psychology

neuroscience

rationality

“human”

biology

sociology

[Hendrycks et al., 2025] [Hutson, 2025]

Defining Intelligence (is really hard)

philosophy

internal processes

behavior

reasoning

mathematics

psychology

neuroscience

rationality

“human”

biology

sociology

[Hendrycks et al., 2025] [Hutson, 2025]

Defining Intelligence (is really hard)

philosophy

internal processes

behavior

reasoning

mathematics

psychology

neuroscience

creativity

rationality

“human”

biology

sociology

[Hendrycks et al., 2025] [Hutson, 2025]

Defining Intelligence (is really hard)

philosophy

internal processes

behavior

reasoning

mathematics

psychology

neuroscience

creativity

rationality

“human”

biology

learning

sociology

[Hendrycks et al., 2025] [Hutson, 2025]

Defining Intelligence (is really hard)

philosophy

abstraction

internal processes

behavior

reasoning

mathematics

psychology

neuroscience

creativity

rationality

“human”

biology

learning

sociology

[Hendrycks et al., 2025] [Hutson, 2025]

Defining Intelligence (is really hard)

philosophy

abstraction

internal processes

behavior

reasoning

mathematics

psychology

neuroscience

creativity

rationality

“human”

biology

learning

sociology

self-awareness

[Hendrycks et al., 2025] [Hutson, 2025]

Defining Intelligence (is really hard)

emotions

philosophy

abstraction

internal processes

behavior

reasoning

mathematics

psychology

neuroscience

creativity

rationality

“human”

biology

learning

sociology

self-awareness

[Hendrycks et al., 2025] [Hutson, 2025]

The most important question to keep in mind when talking about AGI:

The most important question to keep in mind when talking about AGI:

What is the underlying definition of AGI in this context?

The Turing Test

- Indistinguishability in conversation[1]
  - Beat to various degrees
  - Language models “specialized” for this test
- Criticism
  - Can be “faked” easily[2]
  - Language bad indicator for intelligence[3, 4]



[xkcd 329]

[1][Turing, 1949] [2][Weizenbaum, 1966] [3][Fedorenko et al., 2025] [4][The Verge, 2025]

The Turing Test

- Indistinguishability in conversation[1]
  - Beat to various degrees
  - Language models “specialized” for this test
- Criticism
  - Can be “faked” easily[2]
  - Language bad indicator for intelligence[3, 4]

Underlying intelligence understanding:

Indistinguishable (language) between human and AI



[xkcd 329]

[1][Turing, 1949] [2][Weizenbaum, 1966] [3][Fedorenko et al., 2025] [4][The Verge, 2025]

Humanity's Last Exam

- 2500 expert level questions
  - Mostly scientific
  - "Multimodal" (text and images)
- Intended for foundation models
- Traditional LLM benchmarks "too easy"
  - Solution: make the questions harder

Classics

Question:



Here is a representation of a Roman inscription, originally found on a tombstone. Provide a translation for the Palmyrene script. A transliteration of the text is provided: RGYNᵓ BT ḤRY BR ᶜTᵓ ḤBL

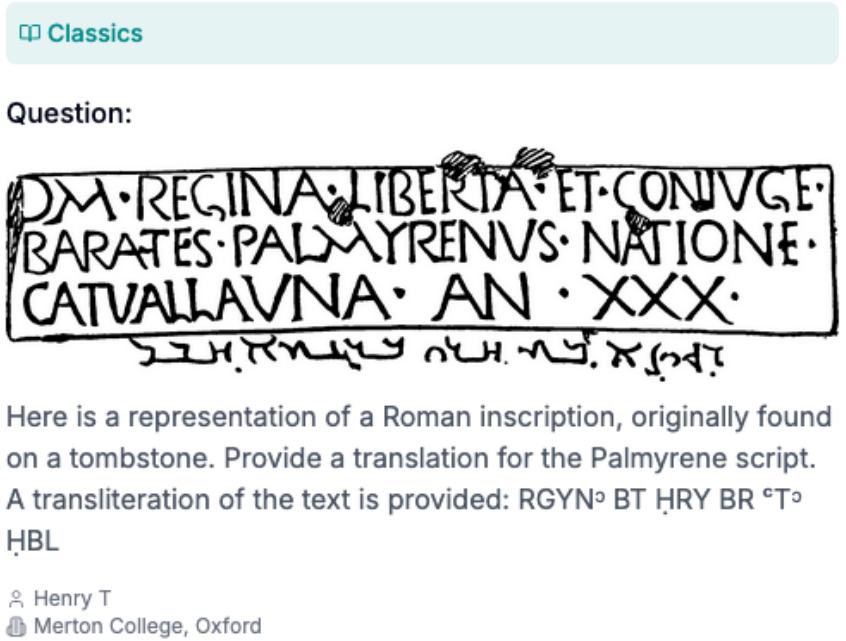
Henry T
Merton College, Oxford

[lastexam.ai]

[Phan et al., 1949]

Humanity's Last Exam

- 2500 expert level questions
  - Mostly scientific
  - "Multimodal" (text and images)
- Intended for foundation models
- Traditional LLM benchmarks "too easy"
  - Solution: make the questions harder



[lastexam.ai]

Underlying intelligence understanding:

Expert level knowledge and reasoning skills across many topics

[Phan et al., 1949]

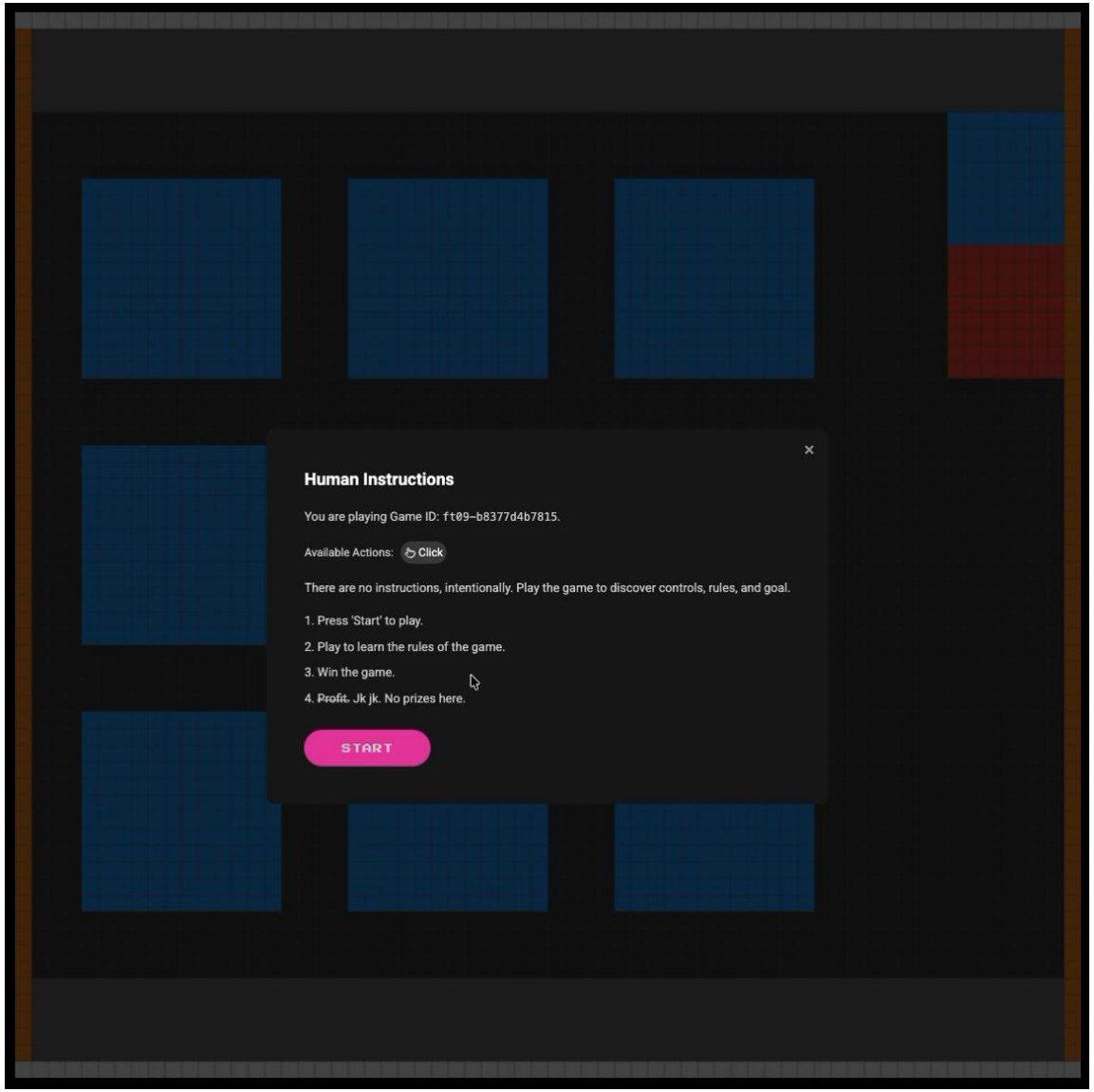
ARC-AGI

- Measuring fluid intelligence
- Static puzzles
  - Somewhat comparable to IQ tests
  - Symbolic, compositional and contextual reasoning
- Interactive puzzles
  - Small games (without given rules)
  - Goal acquisition, planning, exploration

[Chollet et al., 2025] [arcprize.org]

ARC-AGI

- Measuring fluid intelligence
- Static puzzles
  - Somewhat comparable to IQ tests
  - Symbolic, compositional and contextual reasoning
- Interactive puzzles
  - Small games (without given rules)
  - Goal acquisition, planning, exploration



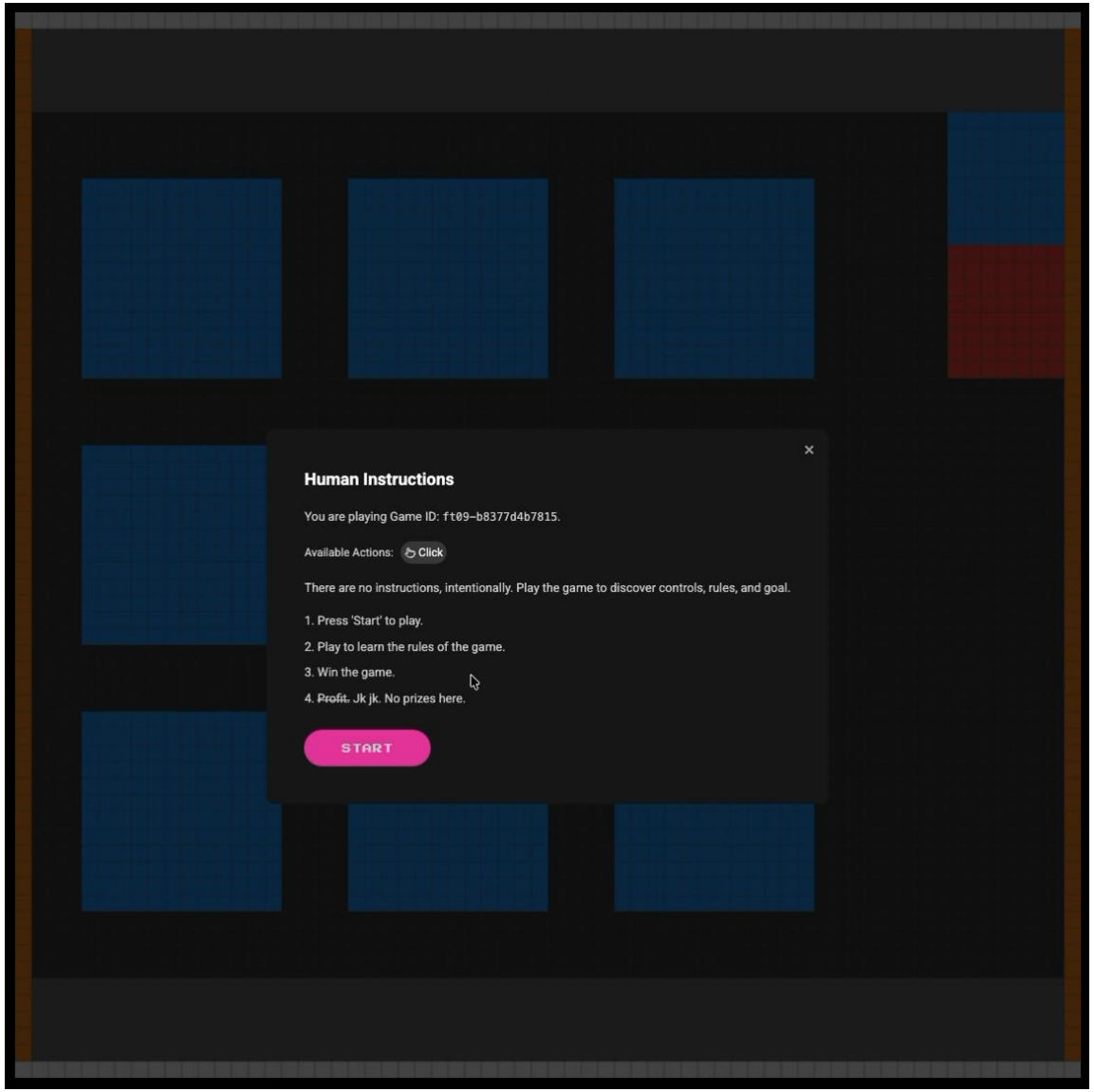
[Chollet et al., 2025] [arcprize.org]

ARC-AGI

- Measuring fluid intelligence
- Static puzzles
  - Somewhat comparable to IQ tests
  - Symbolic, compositional and contextual reasoning
- Interactive puzzles
  - Small games (without given rules)
  - Goal acquisition, planning, exploration

Underlying intelligence understanding:

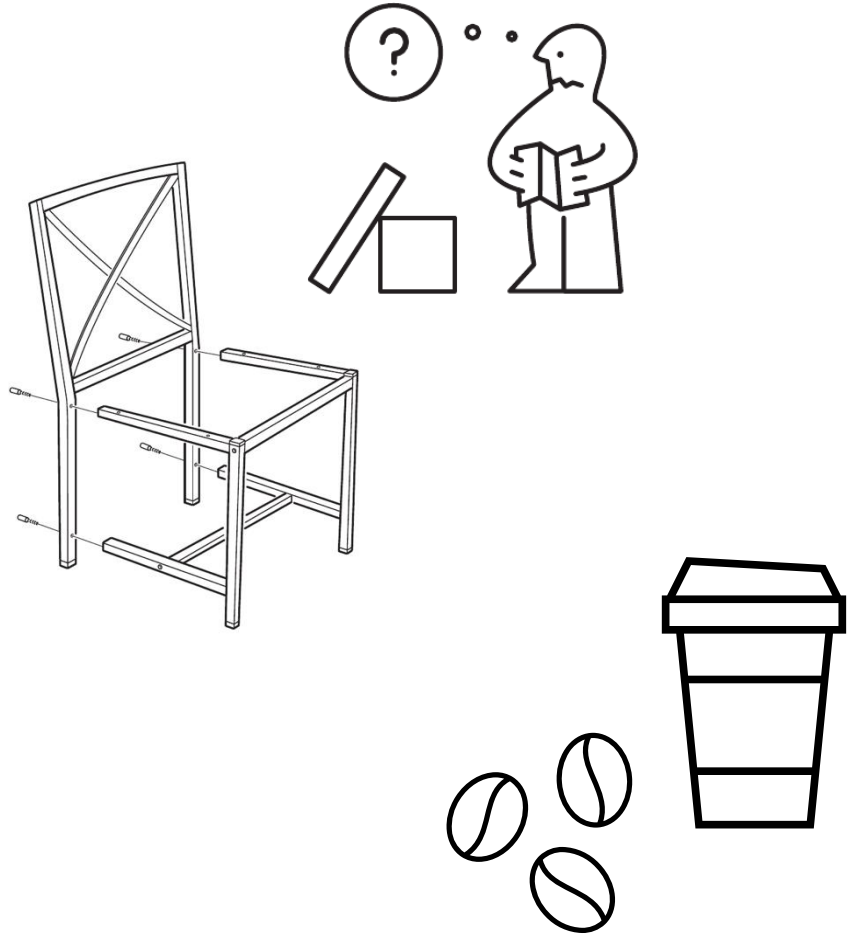
Ability to adapt, learn, and generalize



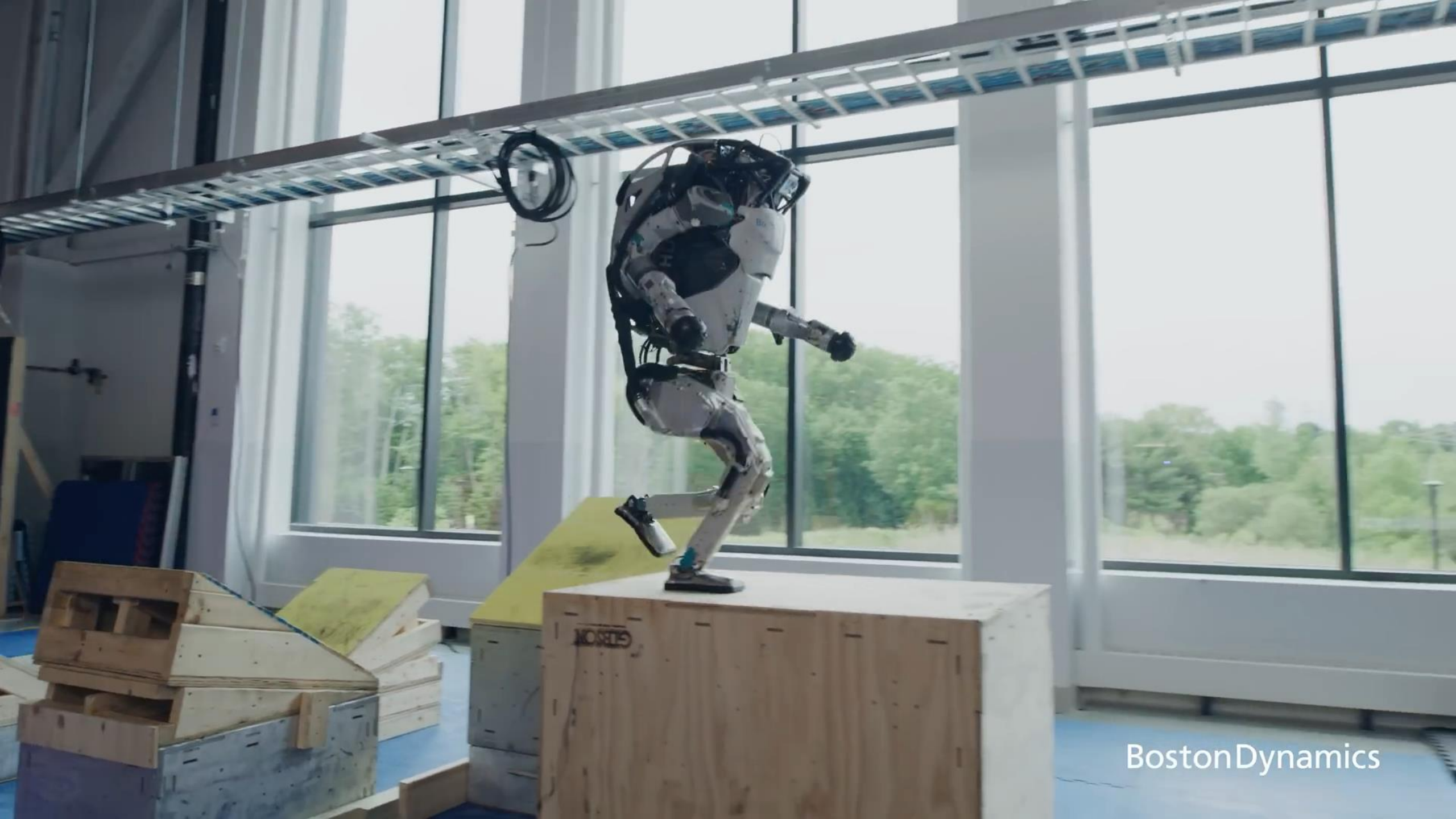
[Chollet et al., 2025] [arcprize.org]

Physical Turing Tests

- Incorporating robotics and embodied AI
- Physical goal fulfillment
  - IKEA test [1]
  - Coffee test [2]
- Physical indistinguishability
  - The Physical Turing Test [3]

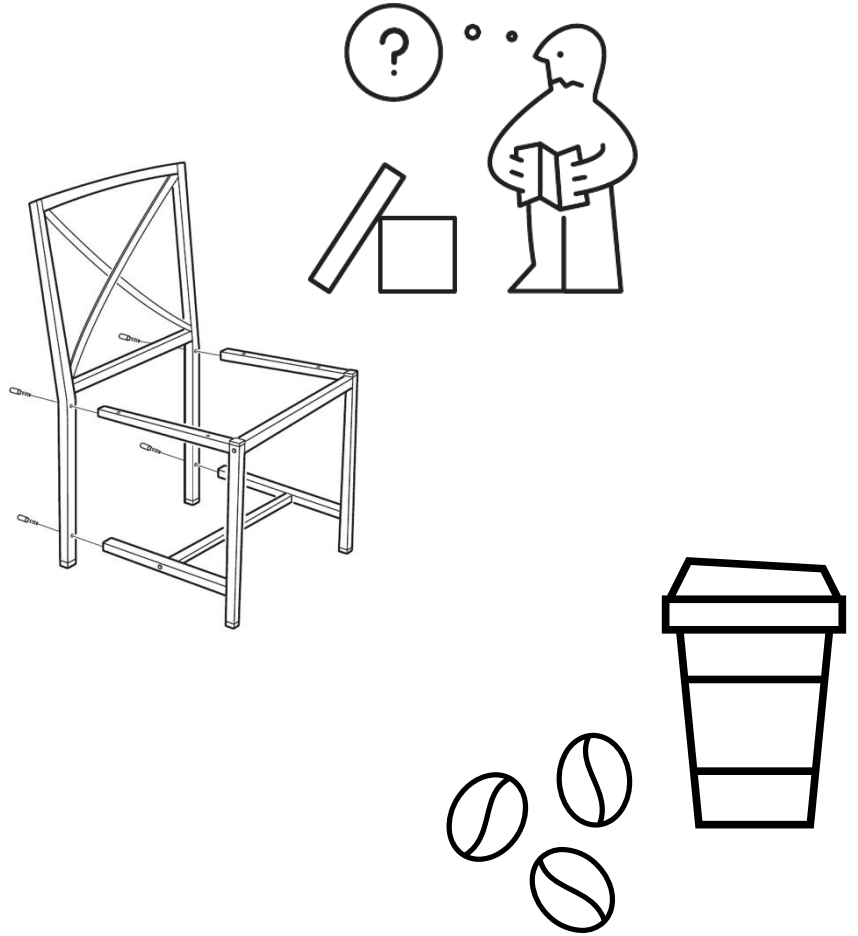


[1] [Vice, 2015] [2] [Goertzel et al., 2012] [3] [Fan, 2025] [Boston Dynamics, 2021]


BostonDynamics

Physical Turing Tests

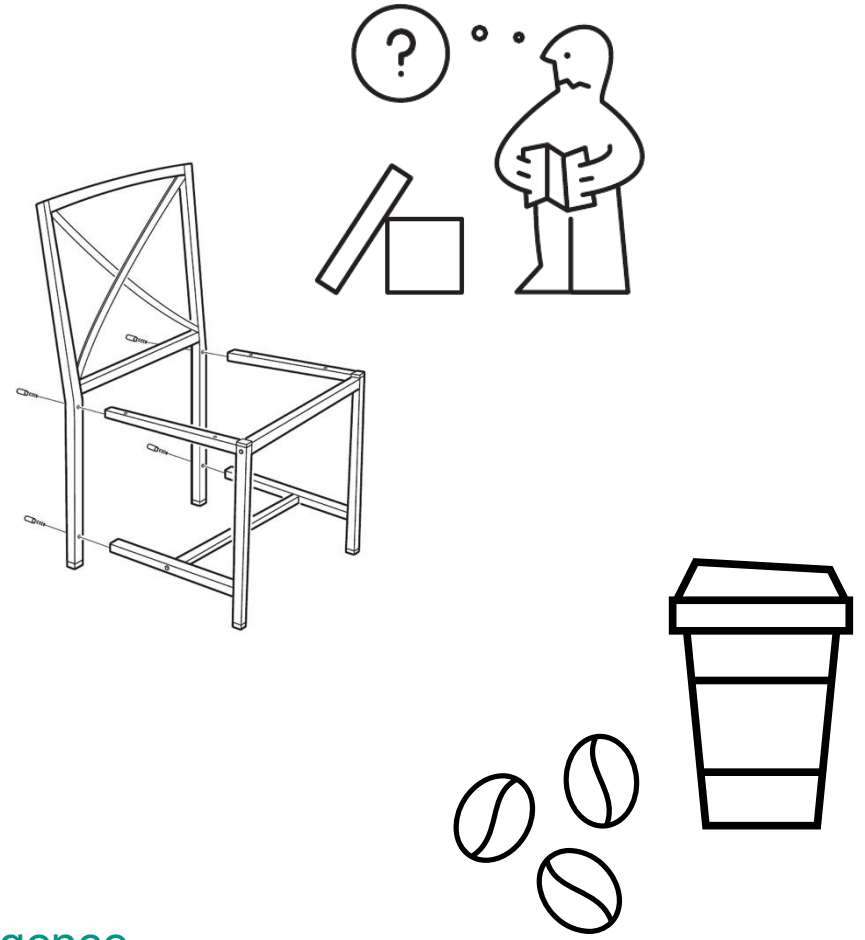
- Incorporating robotics and embodied AI
- Physical goal fulfillment
  - IKEA test [1]
  - Coffee test [2]
- Physical indistinguishability
  - The Physical Turing Test [3]



[1] [Vice, 2015] [2] [Goertzel et al., 2012] [3] [Fan, 2025] [Boston Dynamics, 2021]

Physical Turing Tests

- Incorporating robotics and embodied AI
- Physical goal fulfillment
  - IKEA test [1]
  - Coffee test [2]
- Physical indistinguishability
  - The Physical Turing Test [3]



Underlying intelligence understanding:

Physical presence and interaction as crucial part of intelligence

[1] [Vice, 2015] [2] [Goertzel et al., 2012] [3] [Fan, 2025] [Boston Dynamics, 2021]

Benchmarking AGI

- Many different approaches
- Defined by underlying understanding of intelligence
- Specialized for certain AI subfields/modalities
  - Can be combined
  - But no guarantee for completeness
- No harmonized approach without shared definition

Feasibility - Timelines

When will AGI be reached?

[1][Futurism, 2024] [2][Agüera y Arcas & Norvig, 2023] [3][Futurism, 2023] [4][Business Insider, 2025] [5][Dreyfus, 1992] [6][Fjelland, 2020]

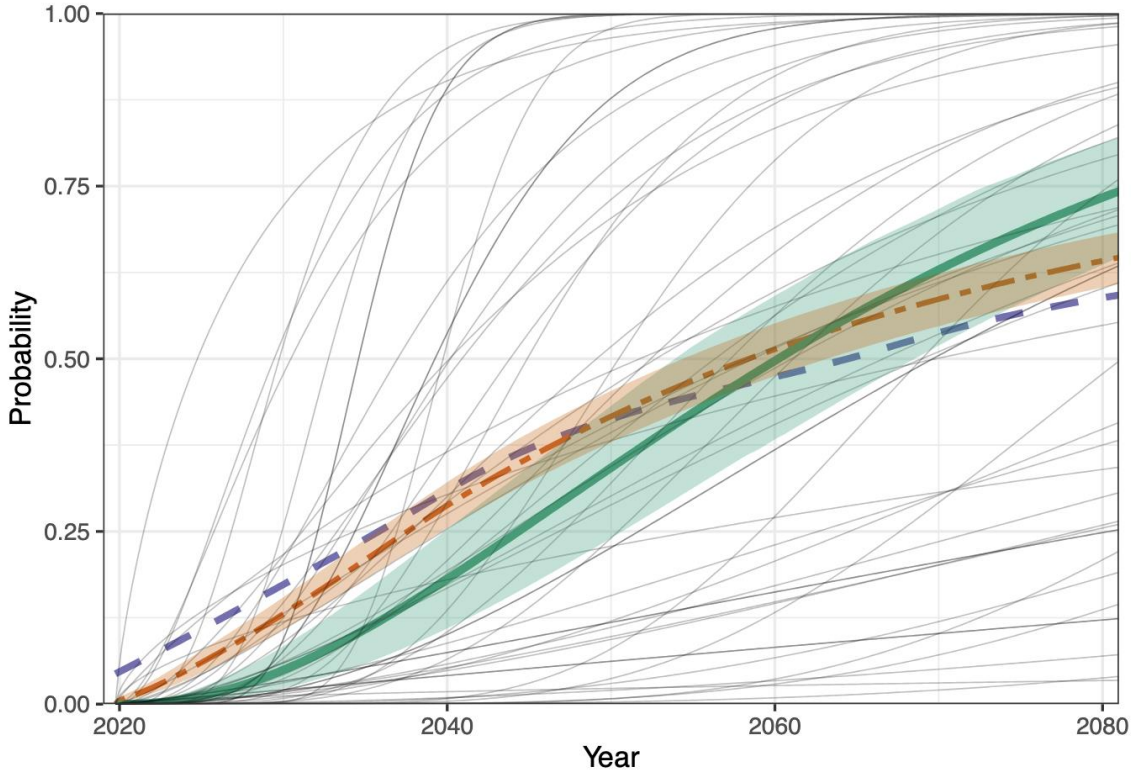
Feasibility - Timelines

When will AGI be reached?



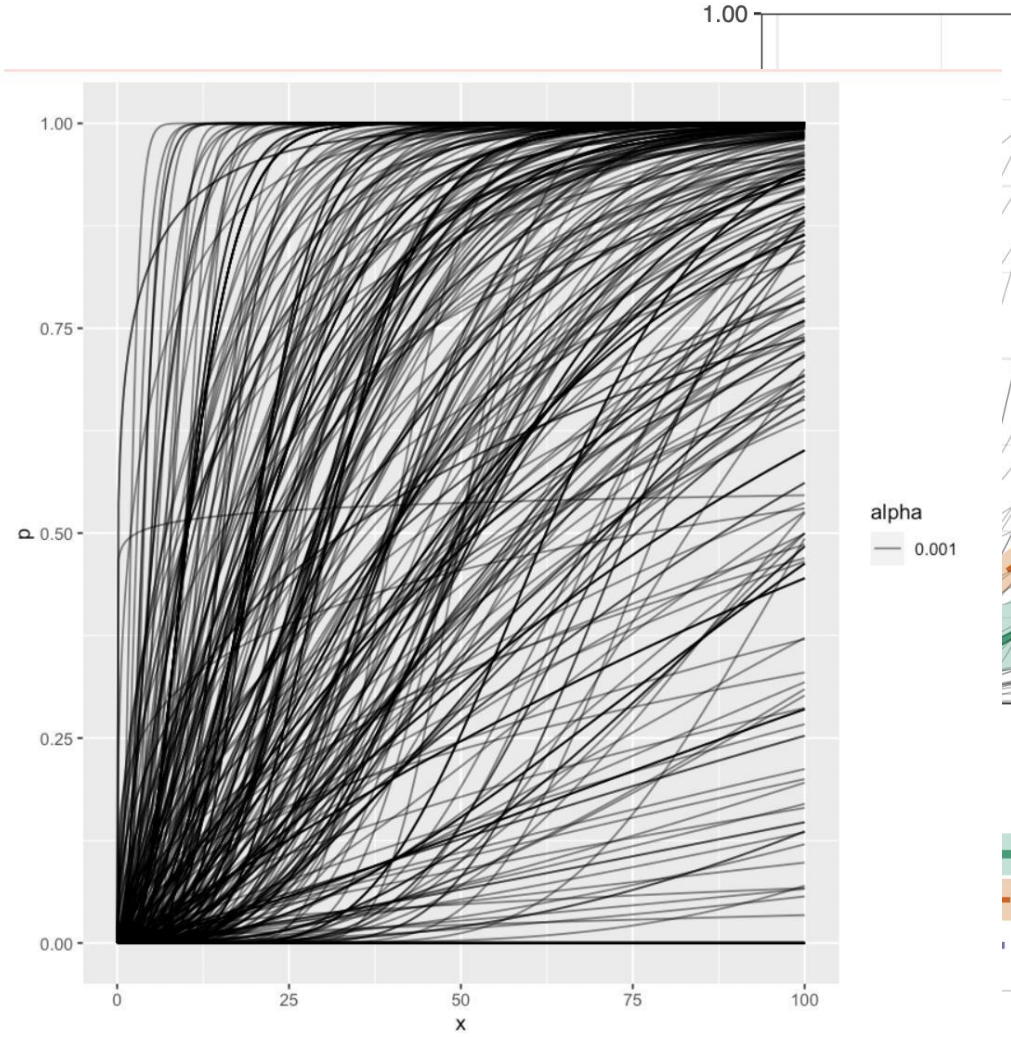
[1][Futurism, 2024] [2][Agüera y Arcas & Norvig, 2023] [3][Futurism, 2023] [4][Business Insider, 2025] [5][Dreyfus, 1992] [6][Fjelland, 2020]

Feasibility - Timelines

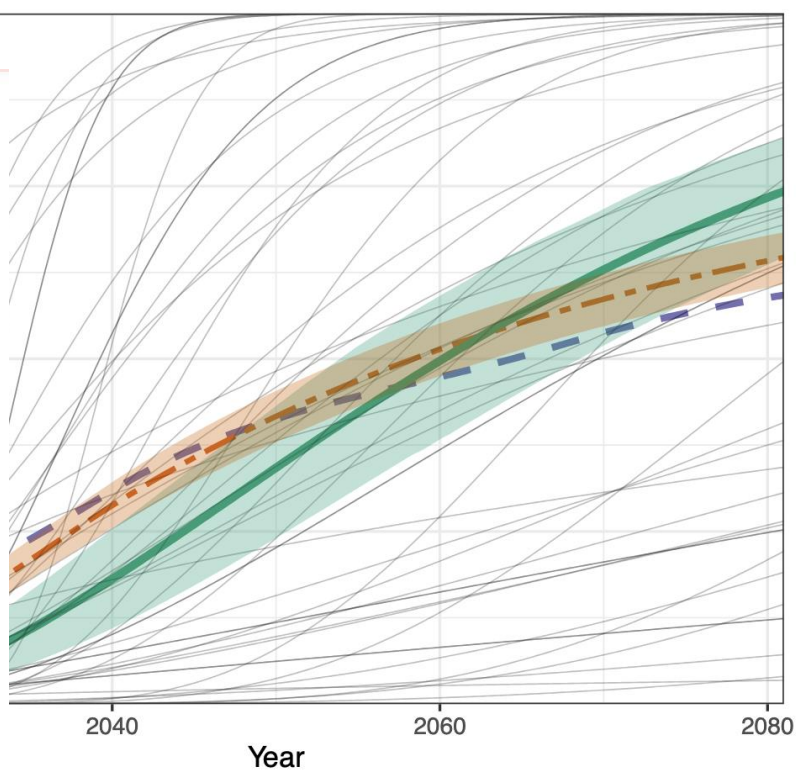


[Zhang et al., 2022]

Feasibility - Timelines

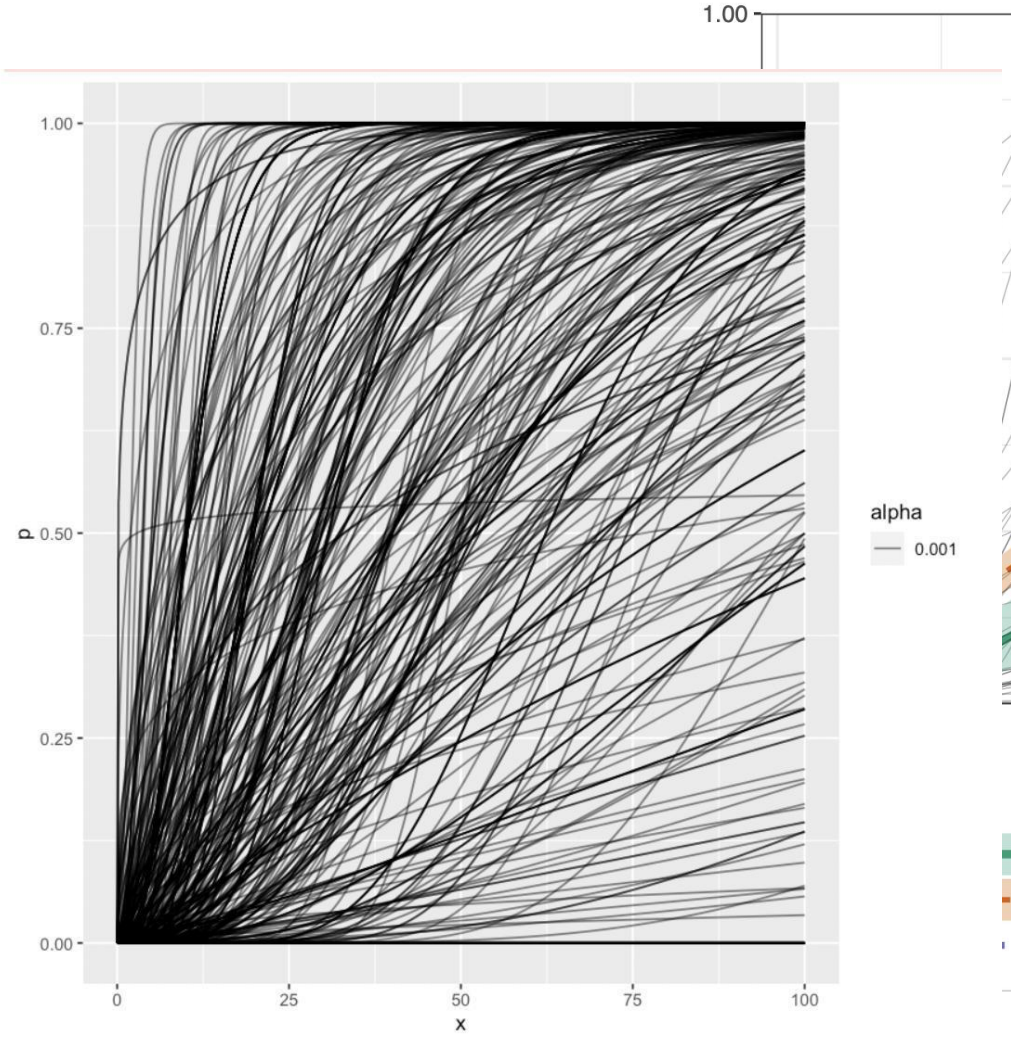


[Grace et al., 2022]

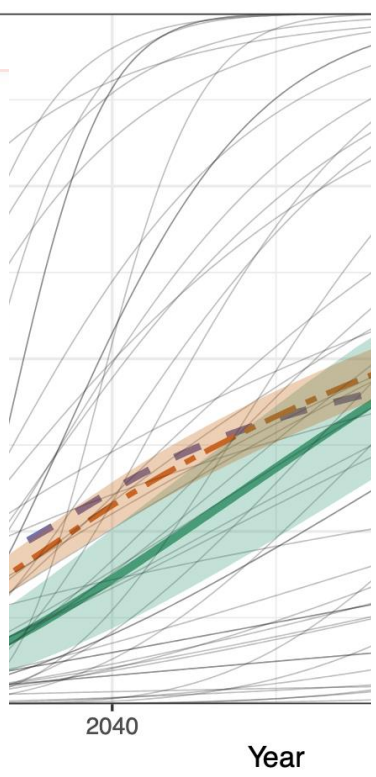


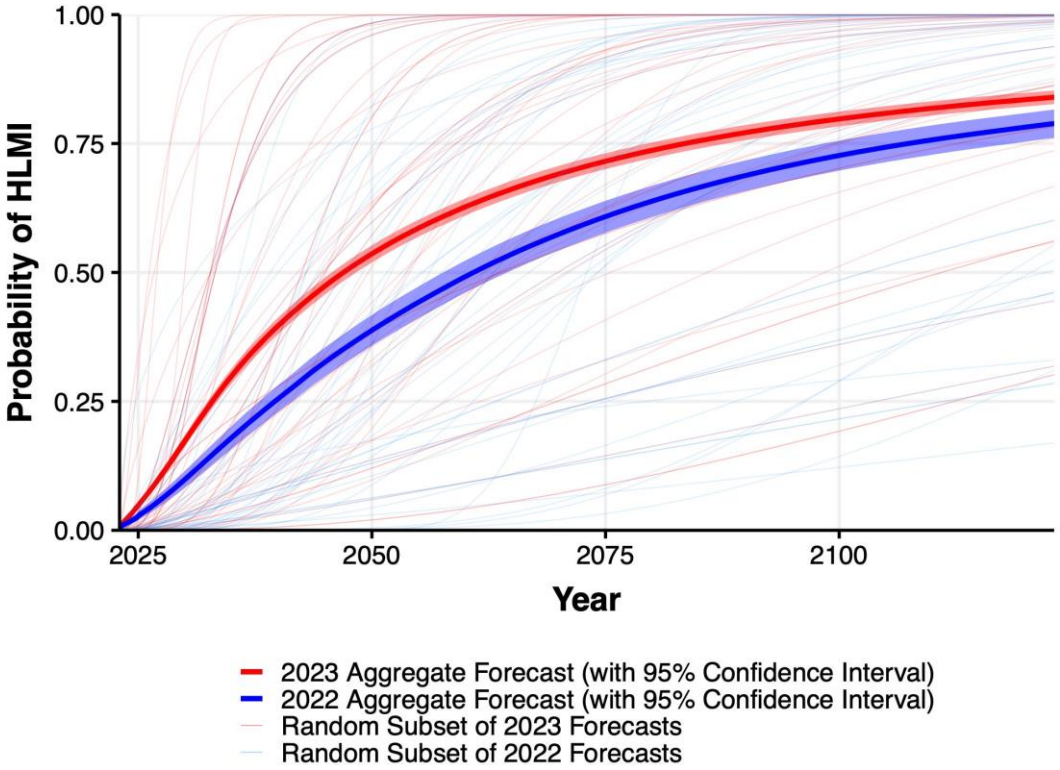
[Zhang et al., 2022]

Feasibility - Timelines

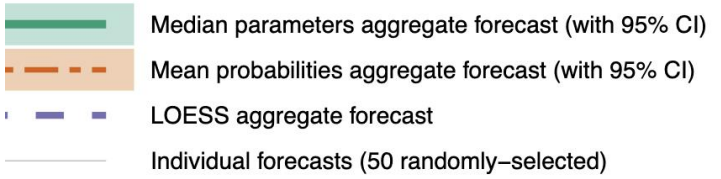


[Grace et al., 2022]



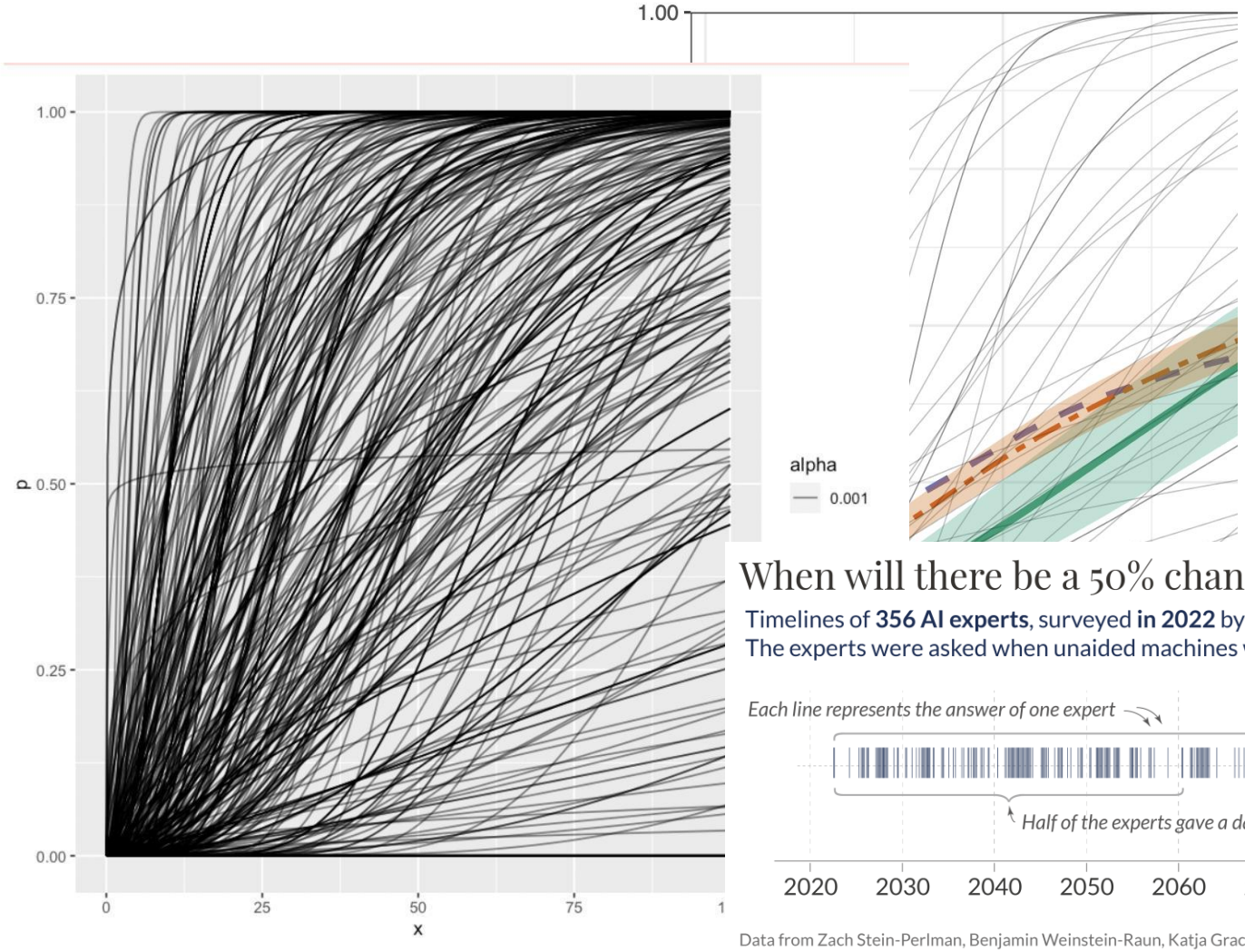


[Grace et al., 2023]

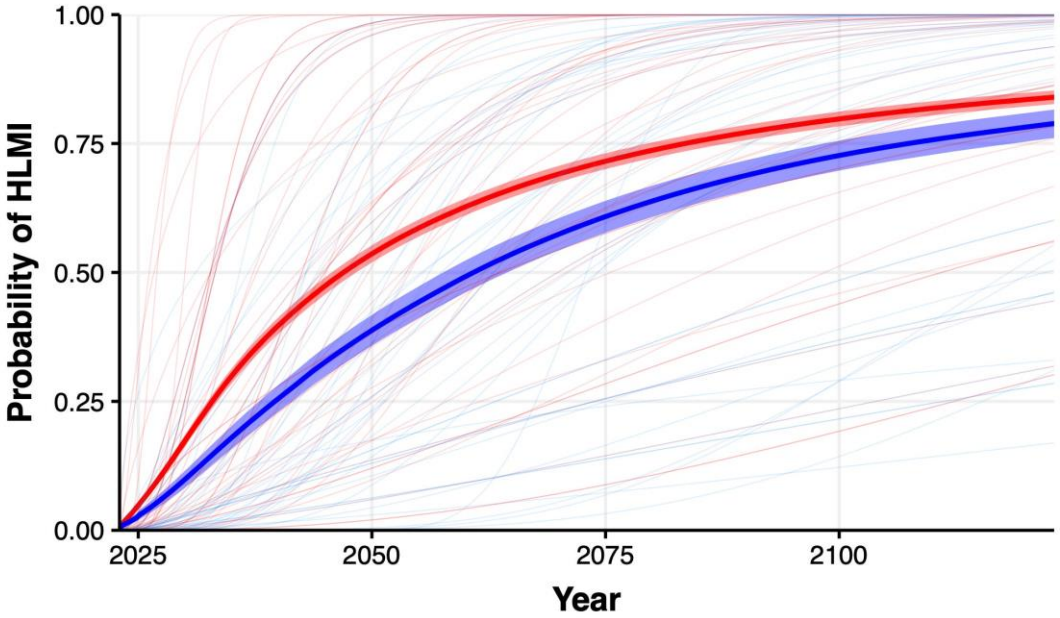


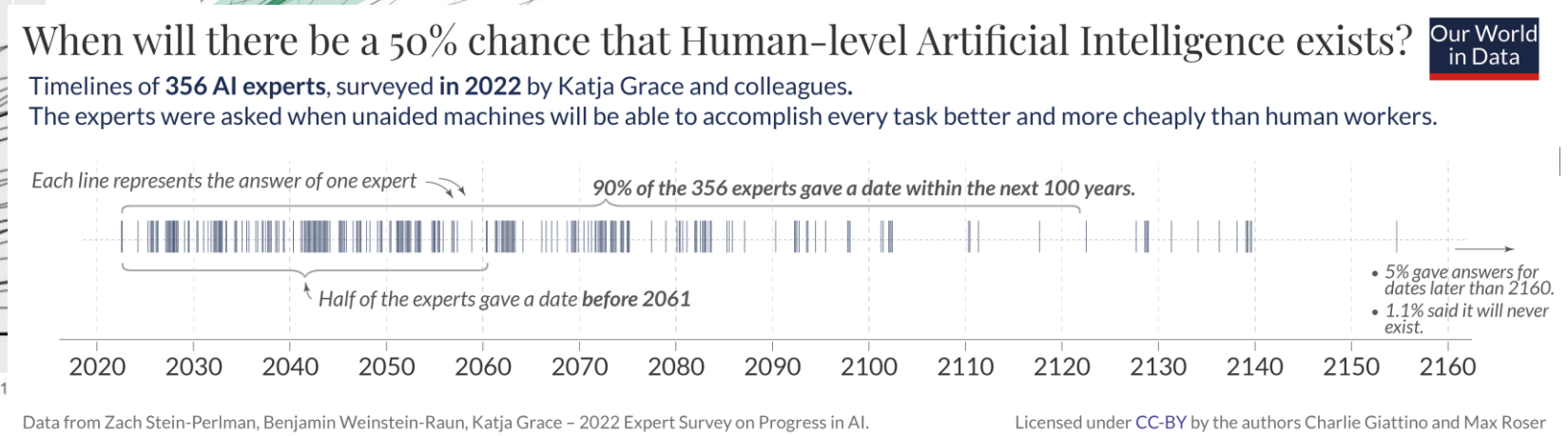
[Zhang et al., 2022]

Feasibility - Timelines



[Grace et al., 2022]





[Roser, 2023]

Feasibility - Timelines

- AGI definitions differ
- “Experts” disagree
- Predictions change

Feasibility - Approaches

1. Advancing SOTA neural networks
2. Biologically inspired AI

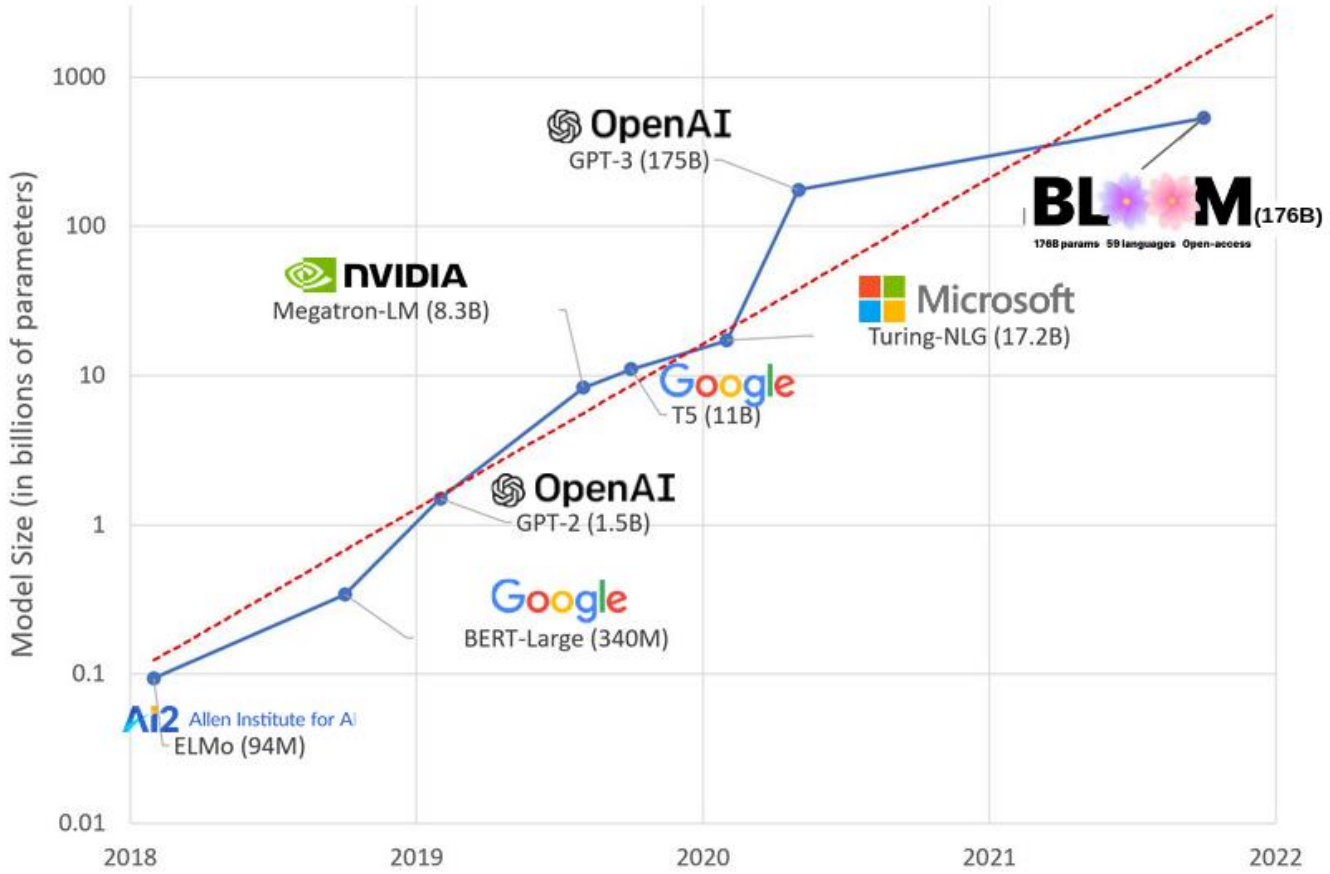
Feasibility - Better NNs

The approach:

- Larger networks
  - More complex architectures
- Increasing modalities[1]
- New learning techniques[2, 3]

The problem:

- Limited training data[4]
- Low data efficiency and generalizability[5]
- No biologic foundation[6]



[ars technica, 2023]

[1][Fei et al., 2022] [2][Baydin et al., 2022] [3][Ouyang, 2022] [4][World Economic Forum, 2025] [5][Huber et al., 2024] [6][Sarfraz et al., 2025]

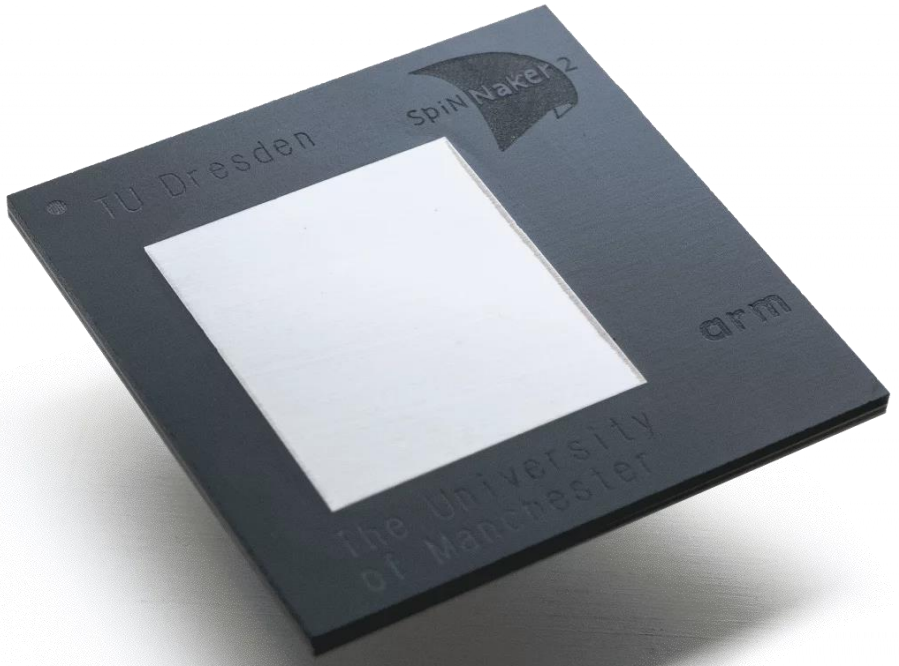
Feasibility - Bioinspiration

The approach:

- Neuromorphic computing
  - Spiking network architectures[1]
  - Specialized hardware[2]
- (Whole) brain emulation

The problem:

- Limited emulation power[3]
- Scanning resolution[3]
- Conceptual correctness[4]



[1] [Maass, 2022] [2] [Furber et al., 2022] [3] [Zanichelli et al., 2022] [4] [Russel & Norvig, 2020]

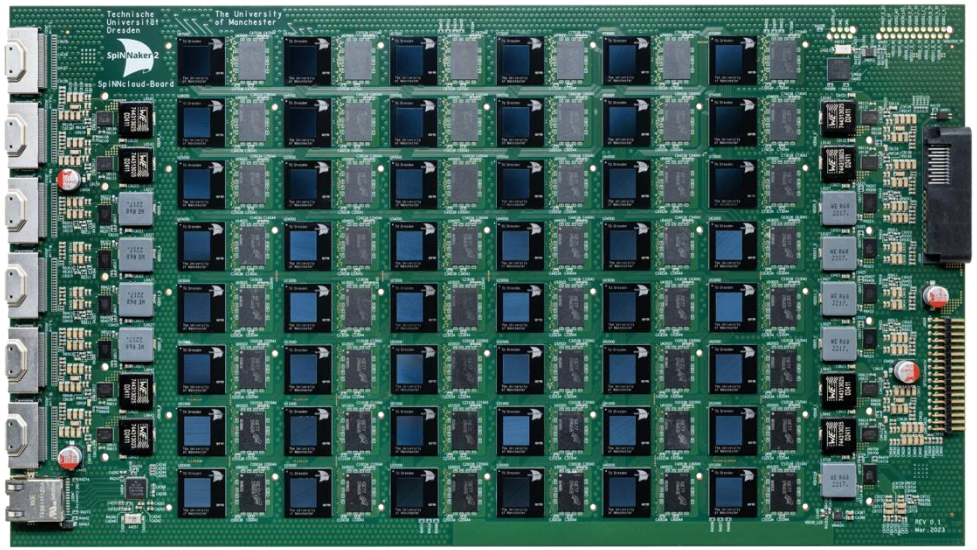
Feasibility - Bioinspiration

The approach:

- Neuromorphic computing
  - Spiking network architectures[1]
  - Specialized hardware[2]
- (Whole) brain emulation

The problem:

- Limited emulation power[3]
- Scanning resolution[3]
- Conceptual correctness[4]



[1] [Maass, 2022] [2] [Furber et al., 2022] [3] [Zanichelli et al., 2022] [4] [Russel & Norvig, 2020]

Feasibility - Bioinspiration

The approach:

- Neuromorphic computing
  - Spiking network architectures[1]
  - Specialized hardware[2]
- (Whole) brain emulation

The problem:

- Limited emulation power[3]
- Scanning resolution[3]
- Conceptual correctness[4]



[SpiNNcloud, Wikimedia]

[1][Maass, 2022] [2][Furber et al., 2022] [3][Zanichelli et al., 2022] [4][Russel & Norvig, 2020]

Implications

“AGI would have a big impact on several aspects of society”

Implications

Governance

Science

Employment

Ethics

Human Creativity

Privacy

“AGI would have a big impact on several aspects of society”

Economy

Security

Social Interaction

Education

Legal Systems

Implications - Prediction Accuracy

- Predictions are hard to impossible
- Increasing layers of assumptions
  - What is intelligence?
  - When is AGI achieved?
  - How does it function?
  - What properties does it have?

Implications - Prediction Accuracy

- Predictions are hard to impossible
- Increasing layers of assumptions
  - What is intelligence?
  - When is AGI achieved?
  - How does it function?
  - What properties does it have?

When encountering predictions ask yourself:

How many and which assumptions are made?

Implications - Prediction Accuracy

- Predictions are hard to impossible
- Increasing layers of assumptions
  - What is intelligence?
  - When is AGI achieved?
  - How does it function?
  - What properties does it have?

When encountering predictions ask yourself:

How many and which assumptions are made?

- Spectrum of predictions
  - Few assumptions -> broad, obvious, boring, likely correct
  - Many assumptions -> specific, interesting, likely incorrect

Implications - Prediction Scale

“AGI would have a big impact on several aspects of society”

Implications - Prediction Scale

“AGI would have a big impact on several aspects of society”



Implications - Prediction Scale

“AGI would have a big impact on several aspects of society”



- Industrial revolution or extinction level event?

Implications - Prediction Scale

“AGI would have a big impact on several aspects of society”



- Industrial revolution or extinction level event?
- Linear, exponential, or disruptive development?

Summary

Summary

- Intelligence is not well understood

Summary

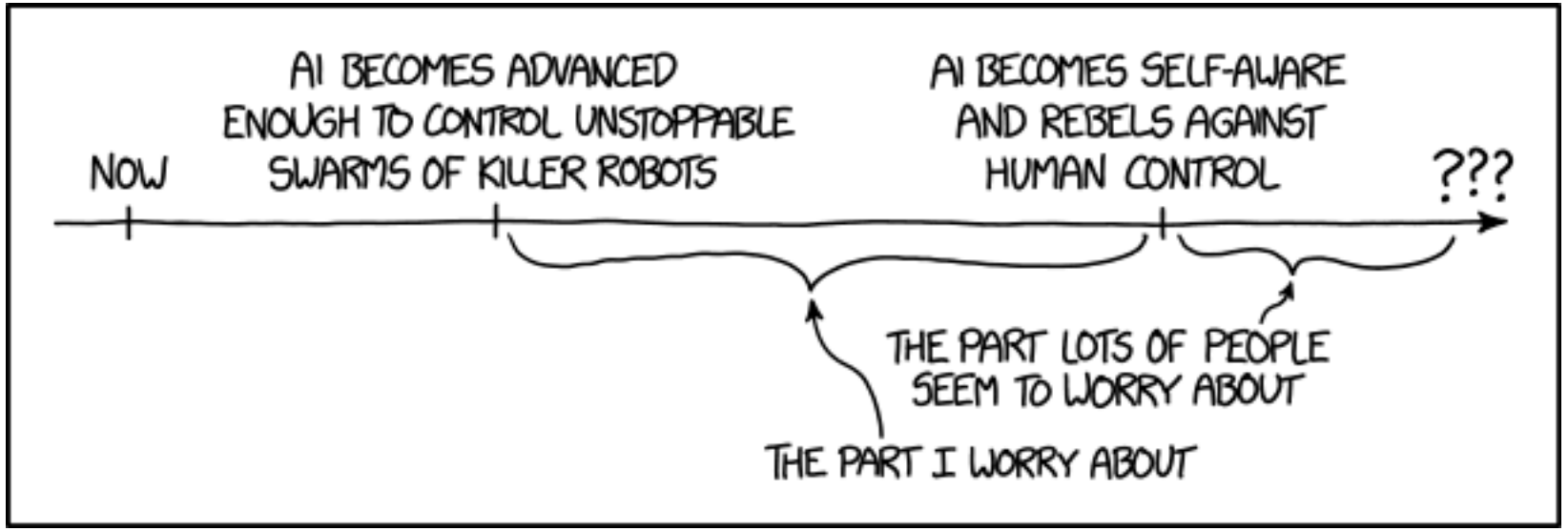
- Intelligence is not well understood
- Ideas and concepts of AGI are diverse

Summary

- Intelligence is not well understood
- Ideas and concepts of AGI are diverse
- This kind of debate is neither new nor useless

Summary

- Intelligence is not well understood
- Ideas and concepts of AGI are diverse
- This kind of debate is neither new nor useless
- Important to focus on current AI developments



[xkcd 1968]

Discussion

- Which properties would you propose for an AGI benchmark?
- Do you consider any current technology a good candidate for a development into AGI?
- What are your thoughts on surveying "experts" on AGI development timelines and feasibility?
- Compared to other societal changes how do you evaluate human-equivalent AGI?

